

Solution Requirements

Date	9 July 2026
Team ID	Not Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	<i>Users (farmers/admin) can securely log in using username and password</i>	Medium
2	Authorization levels	<i>Admin can manage system/data, User (farmer) can input values and get crop recommendation</i>	Medium
3	External interfaces	<i>System connects to crop dataset, ML model, and optionally weather API</i>	High

4	Transactions processing	<i>User inputs (N, P, K, temp, humidity, pH, rainfall) are processed and sent to ML model for prediction</i>	High
5	Reporting	<i>Displays predicted crop result and can show basic recommendation output on screen</i>	High
6	Business rules, etc.	<i>Crop prediction is based on trained ML model using soil and environmental parameters</i>	High
7	Compliance to laws or regulations	<i>Ensure basic data privacy (user input data is not misused or stored without consent)</i>	Low
8	Other	<i>System should allow repeated predictions and easy input modification</i>	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	<i>System should give crop prediction quickly after input</i>	<i>Response time < 2 seconds</i>
2	Scalability	<i>System should handle multiple users using the application</i>	<i>Supports 100+ users simultaneously)</i>
3	Security & Data Privacy	<i>User data should be protected and not exposed</i>	<i>Secure input handling, no data leakage</i>
4	Reliability & Availability	<i>System should work without crashes and be available when needed</i>	<i>95%+ uptime</i>
5	Usability & Accessibility	<i>Interface should be simple and easy for farmers to use</i>	<i>Easy input form, user-friendly UI</i>
6	<i>Other</i>	<i>System should be easy to update or improve</i>	Code should be modular and reusable